

STATUS: READY FOR IMPLEMENTATION

DNK Label Station is a parallel scan-and-print tool under Tools. It reuses the existing warehouse catalog but always prints the retail UPC under the barcode so DNK can match carton labels — without changing the current Label Station short-SKU (Amazon) behavior.

Problem

Warehouse Label Station prints the Amazon short SKU under the barcode when catalog sku has ≤ 7 numeric digits. That ID is correct for Amazon-facing labels, but DNK matches cartons by retail UPC. Staff currently download the PDF and manually replace the short SKU text with the UPC before sending labels to DNK.

Sheets use External ID for the same Amazon workaround; Label Station maps that idea to catalog sku, while upc remains the box barcode.

Goal

Add a separate DNK Label Station that:

- Reuses existing warehouse_products catalog
- Always prints retail UPC under the barcode
- Does not change current Label Station short-SKU rule
- Removes the manual PDF edit step for DNK

Non-goals (v1)

- No new DB table or External ID column
- No Master Sheet / UPC-DIMS changes
- No change to FNSKU Labels tool
- No change to Amazon Label Station rules

Product decisions (v1)

Decision	Choice
Route	/dnk-label-station
Nav label	DNK Label Station (Tools)
Catalog	Shared warehouse_products via existing API
Scan key	Scan / type UPC (same as today)
Under-barcode text	Always product.upc (ignore short-SKU rule)
Barcode payload	Still FNSKU (Mode A)
Catalog import UI	Only on original Label Station
Access	Same as Label Station (requireLabelStationAccess)
Warehouse-only	Add path to WAREHOUSE_ALLOWED_PATHS

Optional later (Mode B): encode UPC in the barcode if DNK must scan the label against the carton.

Data model (unchanged)

Prerequisite: upc must hold the real retail UPC. Short / Amazon ID belongs in sku only. Rows where upc was overwritten with short SKU must be corrected in catalog data.

Field	Role for DNK station
upc	Lookup key + printed under barcode
sku	Amazon short ID — not printed on DNK labels
fnsku	Top text + barcode
style_name / condition	Same as current labels

Label layout comparison

Same physical size (2.25" × 1.25"), sizes small / medium / large, same Zebra / PDF paths.

Region	Current Label Station	DNK Label Station
Top	FNSKU	FNSKU
Barcode	FNSKU	FNSKU
Under barcode	Short SKU if ≤7 digits, else UPC	Always UPC
Right	Condition	Condition
Bottom	Style name	Style name

UX outline

1. Title: DNK Label Station
2. Subtitle: Scan UPC; label prints retail UPC for DNK carton match; same warehouse catalog.
3. Scan row, quantity, Print / Clear / Preview PDF — Enter auto-prints (same model).
4. Label size picker with live preview using DNK renderer.
5. Zebra printer section (Electron) — same desktop print bridge.
6. No import template / upload; note: manage catalog in Label Station.
7. Shared WarehouseProductCatalog browse / select (fills scan with UPC).

Technical design

Print helpers — do not mutate existing short-SKU API

In frontend/src/utils/warehouseLabel.ts (or a thin sibling module): add getDnkLabelScanLine, renderDnkWarehouseLabelCanvas, buildDnkWarehouseLabelZpl, buildDnkWarehouseLabelPdfBlob. Clone / adapt the existing canvas path so the under-barcode line always uses product.upc. Leave getLabelScanLine untouched.

Optional separate localStorage keys so prefs do not collide: dnk_warehouse_label_size, dnk_warehouse_printer_name, dnk_warehouse_printer_dpi.

New page

frontend/src/components/scanner/DnkLabelStation.tsx — forked from LabelStation scan/print shell; wire DNK builders; reuse WarehouseProductCatalog + warehouseProductsApi.lookup; audit events dnk_label_station.print /

dnk_label_station.download_pdf.

Wiring checklist

File	Change
App.tsx	Lazy import + route + ProtectedRoute
Sidebar.tsx	Tools link next to Label Station
NavbarSearch.tsx	Search entry
warehouseAccess.ts	Allow /dnk-label-station for warehouse-only
JobAids.tsx / About.tsx	FAQ: DNK = always UPC; original keeps short-SKU
Backend	None required for v1

Implementation steps

1. Add DNK scan-line + canvas / ZPL / PDF helpers without changing getLabelScanLine.
2. Build DnkLabelStation.tsx from Label Station (strip import UI; swap renderers / copy / audit).
3. Register route, sidebar, search, warehouse allowlist.
4. Add Job Aids / About blurbs.
5. Manual QA on Electron ZPL + browser PDF.
6. (Optional) Unit tests: getDnkLabelScanLine always returns UPC when short SKU is present.

Test plan

- Short-SKU row on current station still prints short SKU
- Same row on DNK station → under-barcode shows UPC; barcode still FNSKU
- Scan UPC → lookup → Enter auto-print (Electron)
- Browser Preview PDF shows UPC text (no hand edit)
- Catalog browse select fills scan and resolves
- Warehouse-only user can open /dnk-label-station
- User without Label Station access is blocked
- Import only on original station; both see same product count after import
- Size / DPI / printer prefs work; do not break original station

Risks

- upc holds short SKU on some rows → data cleanup; document field ownership
- Staff use wrong station → distinct title + sidebar naming
- Mode A insufficient if DNK must scan UPC → Mode B later
- Two large TSX files drift → keep shared logic in warehouseLabel.ts

Success criteria

- DNK labels show retail UPC under barcode without PDF editing
- Amazon Label Station behavior unchanged
- One shared catalog powers both stations

Follow-ups

- Mode B: UPC barcode encoding
- Auto-map sheets External ID → sku
- Vendor filter / DNK-only subset